

UNIVERSITY OF PETROLEUM AND ENERGY
STUDIES

UPES, DEHRADUN

CUSTOMER LIFETIME VALUE PREDICTION AND CUSTOMER SEGMENTATION SYSTEM

A Project Report

Submitted in partial fulfillment of the requirements
for the degree of

Bachelor of Technology

Computer Science and Engineering
Specialization: Data Science

Project Title

Customer Lifetime Value Prediction and Customer Segmentation System

Submitted By

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Declaration

We hereby declare that the project entitled **Customer Lifetime Value Prediction and Customer Segmentation System** is our original work carried out as part of the academic requirements of the Bachelor of Technology programme in Computer Science and Engineering with specialization in Data Science at the University of Petroleum and Energy Studies (UPES).

The project has been developed using machine learning, data analytics, customer segmentation, backend API development, and interactive dashboard technologies.

We further declare that the work presented in this report has not been submitted elsewhere for the award of any other academic qualification.

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Abstract

Customer Lifetime Value (CLV) is an important business metric that estimates the future value a customer can generate for an organization. Traditional customer analysis often focuses only on historical revenue and does not provide a forward-looking estimate of customer value.

This project presents a full-stack machine learning based **Customer Lifetime Value Prediction and Customer Segmentation System**. The system processes customer transaction data, performs data preparation and feature engineering, calculates Recency, Frequency, and Monetary (RFM) features, and uses machine learning to predict customer value.

The project uses an **XGBoost regression model** for CLV prediction and incorporates churn-risk analysis to identify customers who may be less likely to generate future revenue. Customers can subsequently be classified into different CLV tiers such as Low, Medium, and High.

The system provides an interactive web dashboard containing key performance indicators, CLV distribution, revenue forecasting, customer-level predictions, churn-risk information, and customer segmentation.

A custom dataset module is also included. It accepts CSV files in supported RFM or transaction formats and generates customer-level 90-day CLV predictions and churn-risk results.

The trained model achieved an R-squared score of approximately **94.61%**, with an RMSE of approximately **364.93** and an MAE of approximately **111.40** on the evaluated test data.

The resulting system demonstrates how machine learning and business analytics can be integrated into a single application to support customer retention, marketing prioritization, and data-driven decision-making.

Keywords: Customer Lifetime Value, CLV, RFM Analysis, Customer Segmentation, XGBoost, Machine Learning, Churn Prediction, FastAPI, Next.js, MLflow, Customer Analytics.

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1. Introduction

1.1 Background

Customer relationships are one of the most important assets of a business. A company does not only need to acquire customers but also needs to understand how valuable those customers are and whether they are likely to continue purchasing.

Customer Lifetime Value (CLV) provides a way of estimating the expected future value of a customer. Instead of looking only at past purchases, CLV attempts to answer a forward-looking question:

How much future value is this customer expected to generate?

This information can help businesses allocate marketing resources, identify valuable customers, design retention campaigns, and improve customer relationship management.

1.2 Problem Statement

Businesses frequently have large amounts of customer transaction data but lack an integrated system for converting this data into actionable customer-level insights.

A typical transaction dataset may contain hundreds of thousands of records. Manually analysing such data makes it difficult to identify:

- high-value customers,
- customers at risk of churn,
- customers generating repeated revenue,
- customers with declining engagement,
- customers requiring retention campaigns,
- expected future customer revenue.

Therefore, there is a need for an automated system capable of processing transaction data, generating customer-level features, predicting future CLV, and presenting the results through an interactive dashboard.

1.3 Objectives

The main objectives of this project are:

1. To process and clean customer transaction data.
2. To generate customer-level RFM features.
3. To build a machine learning model for CLV prediction.
4. To estimate customer churn risk.
5. To classify customers into CLV tiers.
6. To provide customer-level analytical information.
7. To develop an interactive web dashboard.
8. To support custom CSV datasets through an upload interface.
9. To provide downloadable prediction results.
10. To track machine learning experiments using MLflow.

1.4 Scope

The project focuses on customer transaction analytics and predictive customer value estimation.

The system can be used for:

- e-commerce analytics,
- retail businesses,
- subscription businesses,
- customer relationship management,
- marketing analytics,
- customer retention analysis.

1.5 Project Contributions

The major contribution of this project is the integration of multiple components into one application:

- transaction data processing,
- RFM feature engineering,
- machine learning,
- CLV prediction,
- churn-risk analysis,
- customer segmentation,
- REST API development,
- interactive dashboard,
- custom dataset prediction,
- ML experiment tracking.

2. Literature Review and Background

2.1 Customer Lifetime Value

Customer Lifetime Value represents an estimate of the economic value associated with a customer over a defined future period or across the customer relationship.

A simplified representation can be expressed as:

$$CLV = \sum_{t=1}^T \frac{Expected\ Revenue_t - Expected\ Cost_t}{(1 + r)^t}$$

where:

- T is the considered time horizon,
- r is the discount rate,
- $Expected\ Revenue_t$ is expected revenue,
- $Expected\ Cost_t$ represents expected associated costs.

In this project, the prediction system focuses primarily on expected future customer value over a 90-day prediction horizon.

2.2 RFM Analysis

RFM is a commonly used customer analytics technique.

RFM consists of:

- **Recency:** How recently the customer purchased.
- **Frequency:** How often the customer purchased.
- **Monetary:** How much the customer spent.

For example, a customer who purchased recently, purchased frequently, and spent a large amount is generally more valuable than a customer who has not purchased for a long time.

2.3 Churn Risk

Customer churn refers to a customer becoming inactive or stopping purchases.

A churn-risk score provides an estimate of the likelihood that a customer may become inactive.

In this project, churn risk is used alongside CLV to distinguish customers who are:

- valuable and active,
- valuable but at risk,
- low-value and at risk,
- low-value and relatively active.

2.4 XGBoost

XGBoost is a gradient boosting algorithm based on decision trees.

The model builds multiple weak learners sequentially, with later learners focusing on errors made by earlier learners.

The general objective can be represented as:

$$Obj = \sum_i L(y_i, \hat{y}_i) + \sum_k \Omega(f_k)$$

where L represents the loss function and Ω represents the regularization term.

XGBoost was selected because it performs well on structured/tabular data and can model nonlinear relationships between customer features and future value.

2.5 Machine Learning for CLV

Machine learning allows historical customer behaviour to be used to learn relationships between customer characteristics and future value.

The general workflow is:

Data → Feature Engineering → Model Training → Prediction → Segmentation → Business Insights

3. Dataset and Data Preparation

3.1 Dataset

The primary dataset used in the project is the **Online Retail dataset**.

The original dataset contains approximately:

541,909

transaction records.

The original columns include:

Table 3.1: Original Dataset Fields

Field	Description
InvoiceNo	Invoice number
StockCode	Product stock code
Description	Product description
Quantity	Quantity purchased
InvoiceDate	Transaction date
UnitPrice	Price per unit
CustomerID	Customer identifier
Country	Customer country

3.2 Data Conversion

The original Excel dataset was converted into CSV format for efficient processing.

The generated file was:

`Online_Retail.csv`

The CSV contained:

541,909

rows before transaction filtering.

3.3 Data Cleaning

The ingestion pipeline processes the raw transaction data before storing it in the application database.

The system:

1. reads the CSV file,
2. validates required columns,
3. calculates transaction amount,
4. generates transaction identifiers,
5. generates placeholder email addresses when required,
6. removes invalid transaction records,
7. stores customer information,
8. stores valid transactions.

After processing, the system inserted:

397,884

valid transactions and:

4,338

customers.

3.4 Transaction Amount

Transaction value is derived from quantity and unit price:

$$Amount = Quantity \times UnitPrice$$

This amount becomes an important component of customer monetary value.

3.5 RFM Feature Engineering

The feature engineering pipeline queries transaction information and calculates RFM features for each customer.

The system used:

- Recency,
- Frequency,
- Monetary value,
- Customer tenure,
- Other model-derived features.

The reference date used during the feature engineering process was:

10 December 2011

The resulting feature table contained:

4,338

customer records.

3.6 RFM Statistics

The observed average and median values were:

Table 3.2: RFM Statistics

Feature	Mean	Median
Recency	92.5	51.0
Frequency	90.7	40.0
Monetary	68.35	17.73

These statistics demonstrate that customer behaviour is highly variable, making a machine learning approach useful for identifying patterns.

4. System Architecture and Technical Design

4.1 System Overview

The proposed system follows a full-stack architecture consisting of four major layers:

1. Data Layer
2. Machine Learning Layer
3. Backend/API Layer
4. Frontend/Visualization Layer

4.2 Architecture

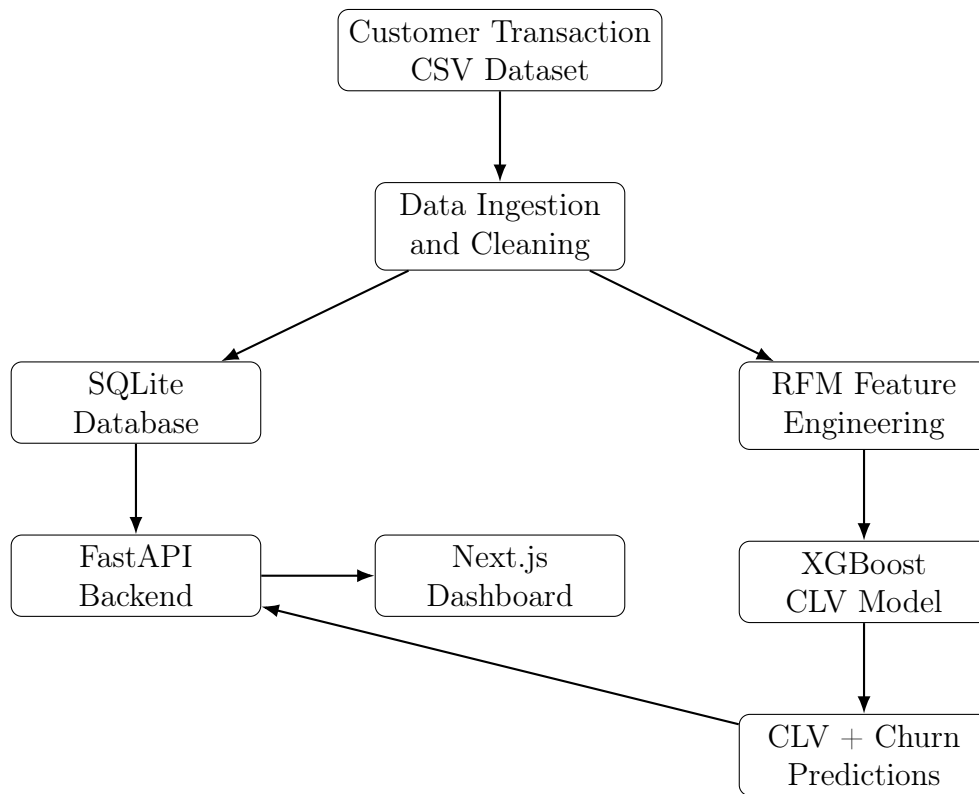


Figure 4.1: High-Level System Architecture

4.3 Technology Stack

Table 4.1: Technology Stack

Technology	Purpose
Python	Data processing and machine learning
Pandas	Data manipulation
XGBoost	CLV prediction model
FastAPI	Backend REST API
Next.js	Frontend dashboard
SQLite	Application database
MLflow	Experiment tracking
Docker	Deployment support
CSV	Dataset input/output

4.4 Backend Architecture

The backend is implemented using FastAPI.

The backend provides APIs for:

- customer data,
- CLV predictions,
- segmentation,
- dashboard statistics,
- custom dataset prediction,
- prediction export.

4.5 Database Design

The system maintains customer, transaction, and feature-related information in the database.

A simplified relationship is:

Customer → Transactions → RFM Features → Predictions

The database allows the application to efficiently retrieve customer records and generate dashboard statistics.

5. Methodology

5.1 Complete Workflow

The complete workflow is:

1. Upload or provide transaction data.
2. Validate the dataset.
3. Clean transaction records.
4. Store customers and transactions.
5. Calculate customer-level RFM features.
6. Prepare the machine learning dataset.
7. Train the XGBoost model.
8. Evaluate model performance.
9. Generate customer CLV predictions.
10. Calculate churn-risk information.
11. Assign CLV tiers.
12. Store or return predictions.
13. Display results through the dashboard.

5.2 Data Ingestion

The ingestion script reads the CSV dataset and prepares it for the database.

An example command used during project execution was:

```
1 python scripts/ingest_data.py "data/Online_Retail.csv"
```

The ingestion process successfully inserted:

- 4,338 customers,
- 397,884 transactions.

5.3 Feature Engineering

The feature engineering process was executed using:

```
1 python scripts/feature_engineering.py
```

The script generated RFM features for all 4,338 customers.

5.4 Model Training

The machine learning model was trained using:

```
1 python scripts/train_model.py
```

The model used XGBoost regression for predicting customer value.

5.5 CLV Prediction

The prediction system estimates expected customer value over a 90-day horizon.

Conceptually:

$$\text{Predicted CLV} = f(\text{Recency}, \text{Frequency}, \text{Monetary}, \text{Tenure}, \text{Other Features})$$

where f represents the trained machine learning model.

5.6 Customer Segmentation

After obtaining predicted CLV, customers are grouped into different value tiers.

A conceptual segmentation is:

Table 5.1: Customer Value Segmentation

Tier	Meaning
High	Customers with high predicted future value
Medium	Customers with moderate predicted value
Low	Customers with relatively low predicted value

The exact threshold logic is implemented by the application rather than being treated as a universal business rule.

6. Machine Learning Model

6.1 Model Selection

XGBoost was selected as the primary CLV regression model because customer transaction data is structured/tabular data and may contain nonlinear relationships.

The model can capture interactions between features such as:

- recent purchasing behaviour,
- purchase frequency,
- monetary spending,
- customer tenure.

6.2 Training Process

The model training pipeline consists of:

Raw Data → Cleaning → Feature Engineering → Train/Test Split → XGBoost → Evaluation

6.3 Evaluation Metrics

The following metrics were used:

6.3.1 R-Squared

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

R-squared indicates how much variation in the target variable is explained by the model.

6.3.2 Mean Absolute Error

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

6.3.3 Root Mean Squared Error

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

6.4 Model Results

The trained XGBoost model produced the following results:

Table 6.1: XGBoost Model Performance

Metric	Result
R-Squared	94.61%
RMSE	364.9261
MAE	111.3960

The R-squared value indicates that the model explains a substantial portion of the variation in the target variable on the evaluated data.

However, model performance should always be interpreted together with the evaluation methodology and business context.

7. Churn Risk Analysis

7.1 Purpose

CLV alone does not completely describe customer behaviour.

For example, a customer may have a high predicted CLV but also have a high probability of becoming inactive.

Therefore, the system presents CLV and churn-risk information together.

7.2 Churn Interpretation

The system can be used to identify combinations such as:

- High CLV + Low Churn Risk,
- High CLV + High Churn Risk,
- Low CLV + High Churn Risk,
- Low CLV + Low Churn Risk.

The most important group for retention campaigns can often be:

High predicted CLV + High churn risk

because these customers may represent significant future value that could be lost.

8. Dashboard and User Interface

8.1 Dashboard Overview

The web dashboard provides an overview of customer analytics.

The dashboard displays key metrics including:

- Average 90-day CLV,
- Active Customers,
- Average Churn Risk,
- Total Revenue,
- CLV distribution,
- Revenue forecast.

8.2 Customer Segments

The Customer Segments page provides customer-level information.

The table can contain:

- Customer ID,
- Email,
- Tenure,
- Total Spent,
- Predicted CLV,
- CLV Tier,
- Churn Risk.

This enables users to move from a high-level dashboard view to individual customer analysis.

8.3 Custom Dataset Prediction

The application also provides a dedicated **Custom Dataset Prediction** module.

A user can upload a CSV dataset.

The system currently supports two input formats.

8.3.1 RFM Format

The RFM format requires fields such as:

```
1 customer_id
2 recency
3 frequency
4 monetary_value
5 tenure
```

8.3.2 Transaction Format

The transaction format requires fields such as:

```
1 customer_id
2 timestamp
3 amount
```

The system processes the uploaded data and generates:

- customer count,
- average predicted 90-day CLV,
- total forecast revenue,
- average churn risk,
- CLV tier,
- customer-level predictions.

8.4 Prediction Export

The custom prediction interface provides an option to export the prediction results as a CSV file.

This allows the user to perform additional analysis outside the application.

8.5 Important System Behaviour

The current implementation separates the standard dashboard data from the custom-upload prediction results.

Therefore:

Uploading a new CSV through the Custom Dataset Prediction page does not currently replace the main Dashboard and Customer Segments dataset.

The custom upload module produces a separate prediction result for the uploaded dataset.

A future version can introduce a unified dataset session so that an uploaded dataset automatically becomes the active dataset for the Dashboard and Customer Segments pages.

9. Implementation

9.1 Project Structure

The project follows a modular structure.

A simplified structure is:

```
1 Customer-Lifetime-Value-Prediction-System/  
2 |  
3 +-- backend/  
4 |   +-- app/  
5 |   +-- data/  
6 |   +-- ml/  
7 |   +-- scripts/  
8 |   +-- .venv/  
9 |   +-- clv_prediction.db  
10 |   +-- requirements.txt  
11 |   +-- Dockerfile  
12 |  
13 +-- frontend/  
14 |  
15 +-- docker-compose.yml  
16 +-- README.md
```

9.2 Data Ingestion Script

The ingestion script is responsible for converting raw transaction data into database-ready records.

The command used is:

```
1 python scripts/ingest_data.py "data/Online_Retail.csv"
```

9.3 Feature Engineering Script

The RFM feature generation is executed using:

```
1 python scripts/feature_engineering.py
```

9.4 Model Training Script

Model training is performed using:

```
1 python scripts/train_model.py
```

The training pipeline also creates an MLflow experiment named:

```
1 CLV_Prediction
```

9.5 Machine Learning Experiment Tracking

MLflow is used to track model experiments and related parameters and metrics.

This improves reproducibility and makes it possible to compare different model runs.

10. Testing and Results

10.1 Data Pipeline Testing

The ingestion pipeline was tested using the Online Retail dataset.

The final processing results were:

Table 10.1: Data Pipeline Results

Item	Count
Original CSV rows	541,909
Valid transactions inserted	397,884
Customers processed	4,338
RFM records generated	4,338

10.2 Model Testing

The model generated:

Table 10.2: Model Evaluation

Metric	Value
R-squared	0.9461
RMSE	364.9261
MAE	111.3960

10.3 Dashboard Testing

The following dashboard components were tested:

- Dashboard KPI cards,
- CLV distribution,
- Revenue forecast,

- Customer segmentation table,
- CLV tier filtering,
- Churn-risk filtering,
- Custom CSV upload,
- Customer-level predictions,
- CSV export.

10.4 Custom Dataset Testing

A sample dataset containing 100 customers was uploaded through the Custom Dataset Prediction module.

The system successfully generated customer-level results including:

- predicted 90-day CLV,
- churn risk,
- CLV tier,
- customer-level RFM information.

This demonstrates that the application is not restricted to the original Online Retail dataset when the uploaded dataset follows one of the supported input schemas.

11. Business Interpretation

11.1 Why CLV Prediction is Useful

The system transforms raw transaction data into business-oriented information.

Instead of asking:

“How much did this customer spend?”

the system helps answer:

“How valuable is this customer expected to be in the future?”

11.2 Marketing Applications

Marketing teams can use CLV predictions to:

- prioritize high-value customers,
- design personalized offers,
- identify retention targets,
- reduce unnecessary marketing expenditure,
- identify customers requiring attention.

11.3 Retention Strategy

Combining CLV and churn risk creates a more useful customer strategy.

For example:

Table 11.1: Example Business Actions

CLV	Churn Risk	Suggested Action
High	High	Priority retention campaign
High	Low	Loyalty and upselling
Low	High	Low-cost retention strategy
Low	Low	Automated engagement

12. Advantages and Limitations

12.1 Advantages

The proposed system provides several advantages:

1. Automated customer feature engineering.
2. Machine learning based CLV prediction.
3. Customer-level segmentation.
4. Churn-risk analysis.
5. Interactive dashboard.
6. Custom CSV prediction.
7. Exportable prediction results.
8. MLflow experiment tracking.
9. Modular backend architecture.
10. Potential for Docker-based deployment.

12.2 Limitations

Despite its capabilities, the current system has several limitations.

1. The primary model is trained using the available historical dataset and may not generalize equally well to every business.
2. The custom upload module currently supports predefined CSV schemas.
3. Uploaded custom predictions are currently displayed in the Custom Dataset Prediction module rather than automatically replacing the main dashboard dataset.

4. The system uses placeholder emails when actual customer email information is unavailable.
5. Predicted CLV should be interpreted as a model estimate rather than guaranteed future revenue.
6. Business-specific costs and profit margins are not fully incorporated into the current CLV calculation.

13. Future Scope

The project can be extended significantly in future versions.

13.1 Unified Dataset Session

The most important improvement is to allow a user-uploaded dataset to become the active dataset for the entire application.

The desired workflow would be:

Upload CSV → Process → Predict → Set as Active Dataset → Dashboard → Customer Segments

This would make the application fully dynamic.

13.2 Explainable AI

Future versions can provide explanations for individual CLV predictions.

For example, SHAP values could explain how:

- Recency,
- Frequency,
- Monetary value,
- Tenure

contributed to a customer's predicted CLV.

13.3 Advanced CLV Models

Future versions may include:

- BG/NBD,

- Gamma-Gamma models,
- probabilistic CLV,
- LightGBM,
- CatBoost,
- neural network based models.

13.4 Real-Time Prediction

The system can be extended to process new transactions continuously and update customer predictions in real time.

13.5 Cloud Deployment

The application can be deployed using:

- AWS,
- Microsoft Azure,
- Google Cloud,
- Docker containers,
- managed databases.

13.6 CRM Integration

Future versions could integrate with CRM platforms so that customer scores automatically trigger marketing campaigns.

14. Conclusion

This project presents a full-stack Customer Lifetime Value Prediction and Customer Segmentation System that combines data engineering, machine learning, backend development, and interactive visualization.

The system processes transaction data, generates RFM features, trains an XGBoost regression model, predicts customer value, evaluates churn risk, and presents customer-level insights through a web dashboard.

The model achieved an R-squared score of approximately 94.61% on the evaluated dataset, demonstrating strong predictive performance for the implemented experimental setup.

The custom dataset module further extends the usefulness of the system by allowing users to upload datasets in supported RFM or transaction formats and obtain customer-level predictions.

Overall, the project demonstrates how machine learning can transform large-scale customer transaction data into actionable business intelligence. The system provides a foundation for customer prioritization, retention analysis, marketing optimization, and future predictive CRM applications.

15. Project Team and Responsibilities

Table 15.1: Project Team

Name	University ID	Role
Chirag Garg	500121978	Project Development
Stuti Singh	500123923	Project Development
Madhav Aggarwal	500123223	Project Development

15.1 Team Contribution

The project was developed collaboratively.

The major areas of work included:

- dataset preparation,
- data ingestion,
- feature engineering,
- machine learning,
- backend API development,
- frontend dashboard development,
- testing,
- documentation,
- project presentation.

A. Project Execution Commands

The following commands were used during project development.

A.1 Virtual Environment

```
1 python -m venv .venv
2 source .venv/bin/activate
```

A.2 Data Conversion

```
1 python -c "import pandas as pd; \
2 df=pd.read_excel('data/Online Retail.xlsx'); \
3 df.to_csv('data/Online_Retail.csv', index=False); \
4 print('CSV created:', len(df), 'rows')"
```

A.3 Data Ingestion

```
1 python scripts/ingest_data.py "data/Online_Retail.csv"
```

A.4 Feature Engineering

```
1 python scripts/feature_engineering.py
```

A.5 Model Training

```
1 python scripts/train_model.py
```

B. Supported Custom Dataset Formats

B.1 RFM Dataset

The RFM dataset should contain the following fields:

```
1 customer_id
2 recency
3 frequency
4 monetary_value
5 tenure
```

An example record is:

```
1 customer_id,recency,frequency,monetary_value,tenure
2 C001,10,5,450.00,365
```

B.2 Transaction Dataset

The transaction format should contain:

```
1 customer_id
2 timestamp
3 amount
```

Example:

```
1 customer_id,timestamp,amount
2 C001,2026-01-10,150.00
3 C001,2026-02-14,250.00
4 C002,2026-01-20,90.00
```

The application uses these transactions to derive customer-level features before prediction.

C. Sample Prediction Interpretation

Suppose the system produces:

Table C.1: Example Customer Prediction

Customer ID	C001
Recency	10 days
Frequency	5 purchases
Monetary Value	\$450
Tenure	365 days
Predicted 90-day CLV	\$520
Churn Risk	20%
CLV Tier	High

This means the model estimates that the customer may generate approximately \$520 of value during the prediction horizon, based on the features available to the model.

It does not mean that the customer is guaranteed to spend exactly \$520.

References

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